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Picrotoxin-induced generalised convulsive seizure in rat: changes in regional distribution and frequency of the power of electroencephalogram rhythms

L. Mackenzie[a], A. Medvedev[a], J.J. Hiscock[a], K.J. Pope[b], J.O. Willoughby[a,*]

[a]*Department of Medicine and Centre for Neuroscience, The Flinders University of South Australia, GPO Box 2100, Adelaide, SA 5001, Australia*
[b]*School of Informatics and Engineering, The Flinders University of South Australia, GPO Box 2100, Adelaide, SA 5001, Australia*

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Abstract

Objectives: It is unknown how generalised discharges in primary generalised epilepsy (PGE) develop from background brain electrical activity or how widespread these discharged are throughout the brain. Here we address this by determining which neural structures and rhythms lead to and participate in generalised discharges in the picrotoxin rat model of PGE.

Methods: Rats with chronically implanted electrodes were infused with picrotoxin until a seizure occurred. This process we refer to as acute epileptogenesis. The electroencephalogram (EEG) was recorded and spectral analysis applied off-line to determine changes in the spectral power of contributing frequencies in 13 brain regions.

Results: Two types of generalised discharge occurred, spindles and seizure, which were present in all brain regions studied. None of the frequencies (1–100 Hz) were significantly increased in background EEG before either spindles or seizure. Within the generalised discharges, power changes revealed significant increases in 6–8 Hz, most powerful in ventrolateral thalamus and neocortex. Gamma frequencies were increased significantly in neocortical structures during spindles with further increases in most structures at seizure onset. 1 Hz was significantly increased in parietal cortex during spindles with differential increases at seizure onset.

Conclusions: We conclude that gamma, 1 and 6–8 Hz frequencies do not appear to contribute to picrotoxin epileptogenesis but do play a role in generalised seizures. The distribution of these frequencies during discharges suggests that the spindles are thalamocortical events and that the seizure is a cortical event with downstream effects on other brain regions. © 2002 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Generalised epilepsy; Picrotoxin; Electroencephalogram; Spectral analysis; Gamma frequencies; Rat

1. Introduction

Primary generalised epilepsy (PGE) is a collective term encompassing several epileptic disorders in human including convulsions. Common to these disorders are discharges with no evidence of localised onset, appearing generalised in a macroscopically normal brain (Porter, 1993). There are few apparent underlying mechanisms for these discharges (Reutens et al., 1993; Niedermeyer, 1996) other than genetic causes (Scheffer and Berkovic, 1997), one of which is a sodium channel abnormality (Wallace et al., 1998). Despite recent developments in identifying genetic abnormalities it is still unknown how seizures emerge from background brain electrical activity. In contrast to PGE, the discharges of partial seizures begin at a focus where they can remain localised or spread to involve other brain regions or to the whole brain, processes which have been extensively studied.

It is generally hypothesised that the increased cerebral excitability associated with seizures is caused by a disturbance in the balance between excitation and inhibition. In this study we use the picrotoxin-treated rat model of generalised convulsive epilepsy (GCE). Picrotoxin reduces inhibition by binding to and blocking the chloride channel associated with the $GABA_A$ receptor when GABA is bound to its specific binding site on this receptor (Newland and Cull-Candy, 1992). Picrotoxin has been used extensively to manipulate the GABA system, both in vivo and in vitro, and to investigate its involvement in epileptic activity (Kaplan and Williamson, 1978; King, 1979; Massotti, 1985; Granata, 2001; Hamani and Mello, 2001). Although the pharmacological properties of picrotoxin are well defined, how seizures develop as a consequence of its action remains elusive.

In the present study, we test the hypothesis that gamma

* Corresponding author. Tel.: +61-618-8204-4689; fax: +61-618-8204-5268.
E-mail address: john.willoughby@flinders.edu.au (J.O. Willoughby).

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frequencies (frequency range 30–100 Hz) are a common feature of the processes leading to ictal events, a process we refer to as acute epileptogenesis. Previously we established gamma frequencies as a persistent feature of kainic acid-induced epileptogenesis in the rat (Medvedev et al., 2000) and postulated that gamma activity might contribute to the epileptic process by synchronising neural activities in different brain structures. Also, it is unknown to what extent brain structures participate in picrotoxin discharges. We aim to establish this distribution. Another type of PGE, the genetically determined generalised non-convulsive or absence seizure has a well-defined thalamocortical substrate (Vergnes et al., 1986; Coenen and van Luijtelaar, 1987; Inoue et al., 1990) in animal models and human. In addition, structures in the Wistar rat brain, other than thalamus and cortex, are differentially involved in these spontaneous spike wave discharges with no discharges recorded from the 'limbic structures' (Vergnes et al., 1990). Here, we accurately define changes in the strength and brain distribution of EEG power induced by picrotoxin using spectral analysis one of the developments in the field of electroencephalography due to the introduction of computer technology.

2. Experimental procedures

To determine the distribution of generalised discharges in the picrotoxin model of GCE we recorded and analysed electrical activity from 12 forebrain and 1 brainstem structures. Examples of EEG are shown in Fig. 1.

2.1. Animals

Thirty adult male inbred Sprague–Dawley rats known not to exhibit spontaneous spike and wave discharges (Willoughby and Mackenzie, 1992) weighing 300–400 g were used. Each animal was anaesthetised with sodium pentobarbitone (60 mg/kg, i.p., Boehringer Ingelheim, NSW, Australia) and implanted with a jugular venous catheter for the infusion of picrotoxin and with various surface or intracerebral stereotaxically positioned monopolar electrodes. Each animal was implanted with bilateral electrodes into 3 different brain structures. In addition an indifferent electrode was placed anteriorly over the frontal sinus and an earth electrode in the occipital bone overlying the cerebellum. Two types of electrodes were used. Stainless steel screws with a flat end diameter of 1.4 mm (Hargo Engineering, Vic., Australia) were used to record from outer cortical structures and also constituted the earth and indifferent electrodes. Intracerebral electrodes consisted of short lengths of 80% platinum-20% iridium 25.4 μm wire (Leico Industries Inc., NY, USA), insulated except for the tip, inside a 30 gauge stainless steel needle. Fine insulated silver wires micro-welded to the steel screws and the intracerebral wires were soldered to a socket into which the EEG recording cable was plugged. The jugular catheter was exteriorised on top of the head via steel tubing and sealed with a plastic cap. The entire assembly was embedded in dental cement forming a robust headpiece.

Although animals were freely moving, eating and drinking within a few hours of the operation they were allowed a standard 4-week recovery period before experimentation. During this time animals were exposed to the recording chamber (30–60 min) on at least one occasion. In all, 13 brain regions were studied; 12 forebrain and one brainstem region. The structures investigated along with their electrode placements and the number of animals with each particular placement are shown in Table 1.

2.2. Experimental protocol

Four weeks after surgery animals were placed in a recording chamber and connected to the EEG recording cable, containing 8 field effect transistor high-input impedance operational preamplifiers (Buzsaki et al., 1988), and an intravenous line. Animals were allowed to settle (around 10 min) then a segment of EEG was collected when the animal was sitting quietly but alert (around 20 s of uninterrupted EEG). Picrotoxin, (Sigma, St. Louis, USA) 1 mg/ml was dissolved in 10% dimethylsulphoxide (Ajax Chemicals, Sydney, Australia), 30 units/ml heparin sodium (David Bull Laboratories, Vic., Australia) and 0.9% saline. The picrotoxin infusion then commenced and was administered via a Sage infusion pump for 1 min every 3 min at a rate of 0.1 ml/min as previously described (Willoughby et al., 1995) until a seizure occurred. The 6 channel EEG signals were recorded throughout the entire experiment with the half amplitude cutoff frequencies set at 0.1 and 1000 Hz, digitised at a rate of 1000 samples per second (MacLab, AD Instruments, Australia) and stored on hard disk using a Macintosh Quadra 950 computer. All experiments commenced between 09:30 and 10:00 h and were video monitored.

2.3. Electrode placement verification

Immediately following the first seizure each animal was deeply anaesthetised with sodium pentobarbitone, transcardially perfused and the brain removed. Vibratome sections (50 μm) were slide-mounted and stained with cresyl violet for microscopic verification of electrode placement (Paxinos and Watson, 1986). Only animals with a full complement of accurately placed electrodes were included in the analysis.

2.4. Spectral analysis

The spectral content of the EEG signals was determined by off-line analysis using a programme written using Igor software (WaveMetics, USA) as previously described (Medvedev et al., 1996). To quantitatively determine changes occurring in EEG activity at different time points we performed spectral analysis on the following EEG segments for each animal:

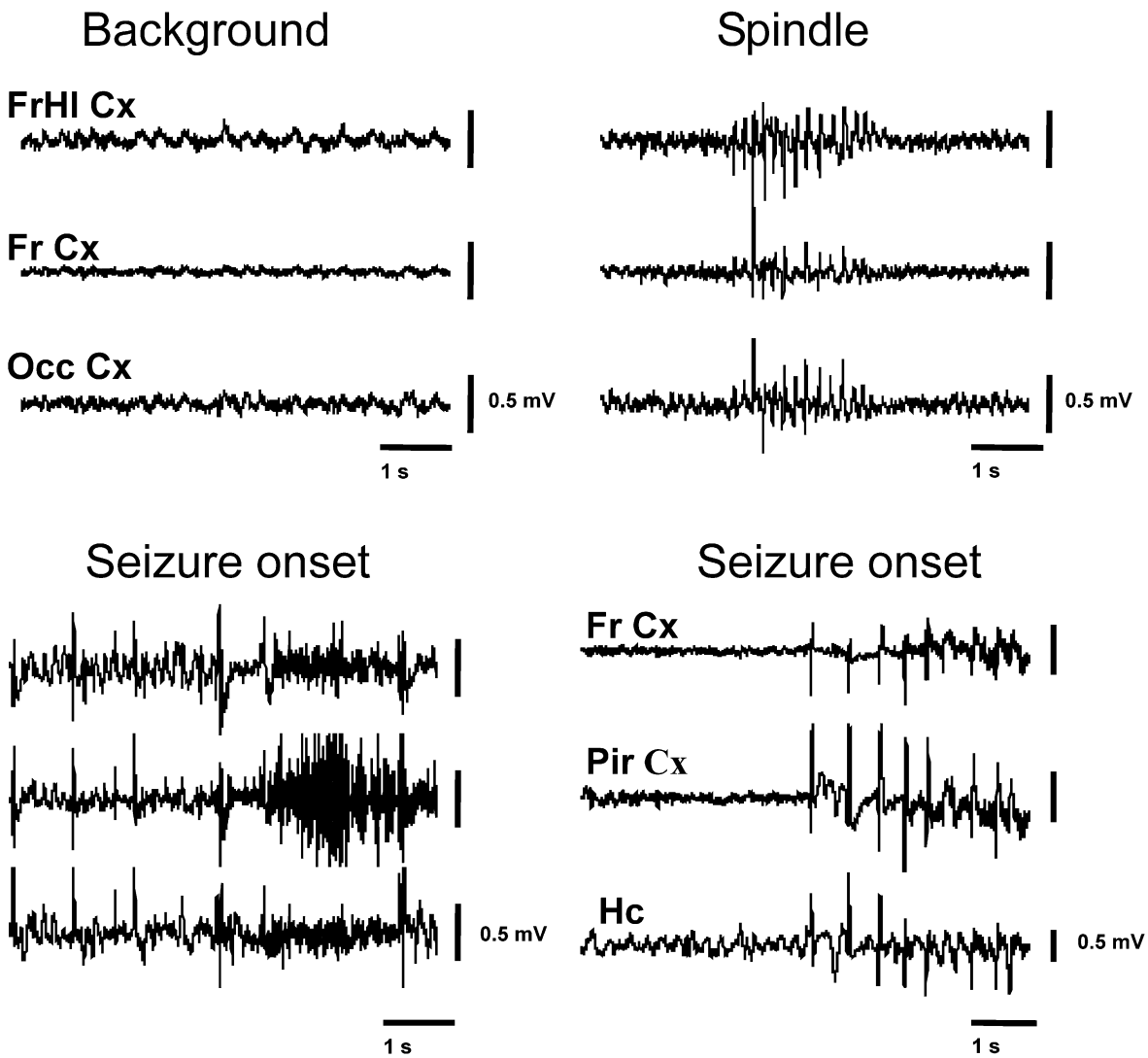

Fig. 1. Examples of multichannel EEG from left side of the brain in one animal showing: (a) background; (b) spindle; (c) seizure onset immediately preceded by spindling activity; and (d) seizure onset immediately preceded by background EEG activity in a different animal (different electrode configuration). ↑ marks seizure onset. Abbreviations: FrHl Cx = frontal hindlimb cortex, Fr Cx = Fr1 area of motor cortex, Occ Cx = occipital cortex, Pir Cx = piriform cortex, and Hc = hippocampus.

1. Segment before any drug was given (control segment) (10 s). Animals were quiet but alert and moving around.
2. Segment of background activity following drug but before the first generalised high voltage spike-and-wave EEG pattern associated with myoclonic jerk (spindle) (4–6 s).
3. Segment of background activity as close to the seizure as possible (4–6 s).
4. Initial spindle (2–4 s).
5. Last spindle (2–4 s).
6. Beginning of the seizure (seizure onset) (2 s).

All segments were analysed by dividing them into half-overlapping 1.024 s epochs which were weighted by a Hamming spectral window and Fast Fourier Transformed. The smoothed power spectrum of a segment was calculated for 1–100 Hz as an average of all epochs. The spectra obtained from segments 2–6 were expressed relative to the control segment and the ratio plotted as a relative spectral response (Fig. 2). For each structure being investigated the relative responses for all rats in the group were averaged. Confidence limits for averaged relative responses were calculated at the level of 0.01 using F ratios determined by the number of animals. Relative spectral responses were also calculated to determine if initial spindles (segment 4) had the same spectral content as the last spindles (segment 5) and also if the spindle frequencies (segment 5) were altered at the beginning of the seizure (segment 6).

2.5. Statistical analysis

The power of frequencies representative of spindles and gamma (extracted from the relative spectral responses) at

Table 1
Areas implanted with electrodes and their numbers and relative positioning in mm (Paxinos and Watson, 1986)[a]

Area	*N*	Anterior-posterior to bregma	Lateral to midline	Depth below brain surface
FrHl Cx	18	−2.0	2.5	Surface
Fr1	16	+1.2	2.0	Surface
Par Cx	3	−2.3	4.5	Surface
Occ Cx	6	−6.0	2.5	Surface
Pir Cx	3	−0.4	4.6	8.6
Ento Cx	4	−6.3	4.5	8.0
Cing	7	+3.2	0.5	3.0
VL Thal	4	−1.8	2.0	6.0
Re Thal	3	−1.4	1.6	6.6
PO Thal	3	−3.3	2.0	5.5
Hc	7	−3.8	2.0	3.4
Amyg	3	−3.3	4.5	8.6
PnO	6	−8.8	1.0	8.0

[a] Abbreviations; FrHl Cx = frontal hindlimb cortex, Fr1 = Fr1 area of motor cortex, Par Cx = parietal cortex, Occ Cx = occipital cortex, Pir Cx = piriform cortex, Ento Cx = entorhinal cortex, Cing = cingulate cortex, VL Thal = ventrolateral thalamus, Re Thal = reticular thalamic nucleus, PO Thal = posterior thalamic nuclear group, Hc = hippocampus, Amyg = amygdala, PnO = pontine reticular nucleus-oral. *N* = the number of animals with bilateral electrodes implanted in this area. EEG data was collected to also allow determination of temporal relationships between structures (to be published elsewhere). This meant that certain structures appeared in several groupings of electrodes hence the unequal numbers reported here.

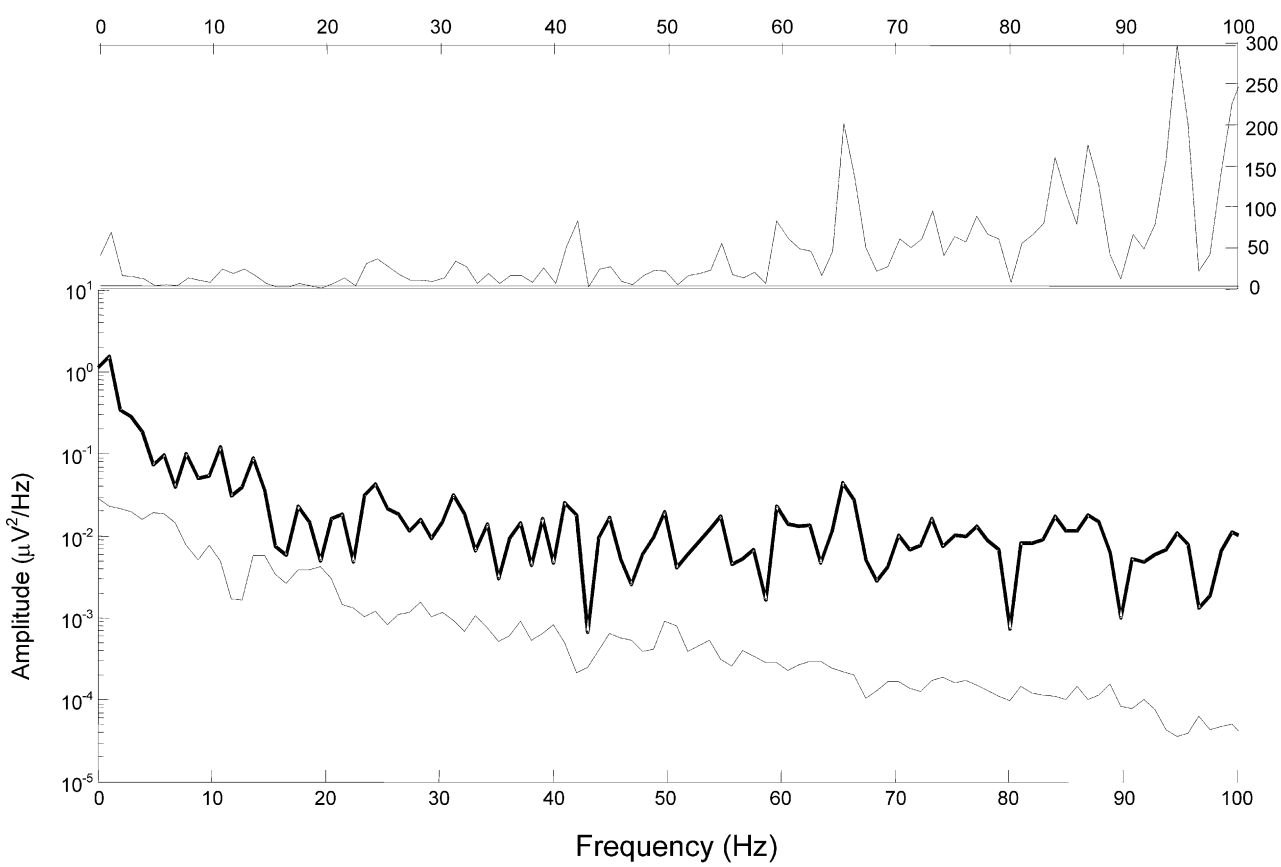

Fig. 2. Determination of relative responses. Lower graphs show power spectra plotted for pre-drug background segment of EEG (thin line) and for a response segment of EEG, in this case for seizure onset (thick line). These spectra are from one rat from frontal hindlimb cortex on the left side of the brain. Upper graph shows the relative response, i.e. the ratio of the two spectra. The broken lines indicate the upper and lower confidence limits calculated by *F* ratio at 0.01 significance level.

different stages of the experiment was compared for each brain region. Data was first log transformed to meet the normal distribution assumptions of analysis of variance (ANOVA) and then analysed using a general linear model ANOVA. This was followed by post hoc analysis using Tukey's pairwise comparisons (Minitab Statistical Software, PA, USA). A significance level of 0.05 was used for these analyses.

3. Results

As previously reported, picrotoxin caused two types of discharges, EEG spindles associated with myoclonic jerks (Kaplan and Williamson, 1978; King, 1979; Ben-Ari et al., 1981; Massotti, 1985; Medvedev et al., 1996) and discharges associated with seizure. By visual inspection, discharges were truly generalised in that they appeared simultaneously in all brain regions studied although they were not expressed to the same extent. All animals exhibited multiple spindles separated by periods of normal background EEG before a seizure occurred. In addition, there were distinct episodic events confined mainly to the frontal hindlimb cortical electrodes. These events were of short (around 1–2 s) duration, of low voltage and exhibited a peak frequency of 12 Hz. They appeared before the initial spindle occurred and were present between subsequent spindles. We interpreted these events to be rhythmic sensory-

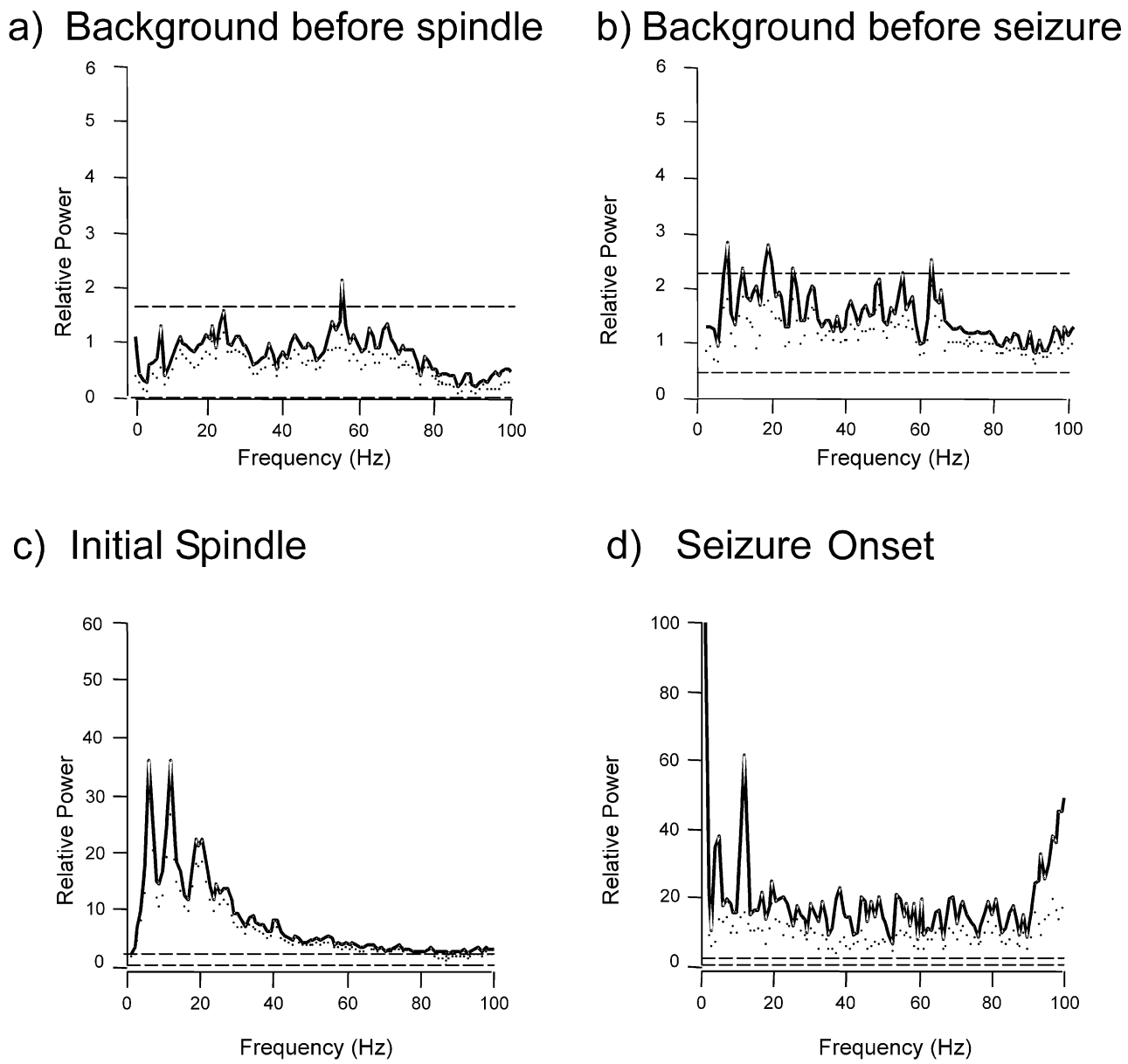

Fig. 3. Examples of average relative responses for segments of EEG recorded from frontal hindlimb cortex, left side only, $N = 18$. The broken lines indicate the upper and lower confidence limits calculated by F ratio at 0.01 significance level. A level of 1.0 = no change. Dots at each frequency indicate the lower SEM.

motor or mu oscillations on the basis of their distribution and dominant frequency. The spectral profile for 10 spindles selected at equal intervals from the initial spindle to seizure onset were very similar (data not shown) therefore, only the first and last spindles were used in the analysis. Spindles did however become more frequent throughout the epileptic process. By visual inspection, EEG activity immediately preceding a seizure was normal in 13 of the 30 animals while there were spindles in the remaining 17. Animals required 1.5 ± 0.21 mg/kg of picrotoxin to cause a seizure (range: 1.07–1.91mg/kg).

3.1. Spectral analysis

There was no significant increase in power at frequencies 1–100 Hz in the background activity of any brain structure, as shown by the averaged relative spectral responses (Figs. 3a,b). This was the case regardless of whether the background segments were taken before the initial spindle (segment 2) or as close to seizure as possible (segment 3). Single frequencies sometimes exceeded the significance level but there were large variances as demonstrated by the error bars. Changes in the relative spectral responses of certain frequencies in the alpha and gamma wavelength bands occurred during discharges. These frequencies, namely 1, 6–8 and 30–100 Hz, were investigated fully. There is very little, if any, electromyographical contamination of the EEG signal below 100 Hz as determined by experiments we have undertaken in paralysed rats given picrotoxin, unpublished.

Spindles were present in every brain region from which we recorded. Within every spindle there was an increase in the power of frequencies at 6–8 Hz relative to background activity. This dominant peak as well as probable harmonics of 6–8 Hz frequencies resulted in a tri-phasic-peak appearance in the power spectrum (Fig. 3c). A 6–8 Hz peak was also present at seizure onset (Fig. 3d). In order to compare the relative power of oscillations at the spindle frequency, the strength of the dominant peak from 6 to 8 Hz was used, ignoring the contribution to power of probable harmonics. This avoided the impossible task of separating the power due to the harmonics from that of endogenous oscillations at each frequency. The relative power of the dominant peak differed significantly throughout the experiment (segments 2–6, $P < 0.001$, Fig. 4b). A significant increase occurred in the relative power of 6–8 Hz contributing to the initial and last spindles ($P < 0.001$ for both) but not in the preceding background. Initial and last spindles demonstrated similar strengths of 6–8 Hz oscillations, and both were stronger than at the beginning of the seizure ($P < 0.001$, $P < 0.01$ for initial and last spindles respectively). There was differential expression in different brain regions of spindle power ($P < 0.001$), with ventrolateral thalamus followed by neocortex demonstrating the most power at this frequency and with amygdala and cingulate cortex having least power (Fig. 5b).

The relative power of gamma frequencies (30–100 Hz) differed significantly through the experiment (segments 2–6, $P < 0.001$, Fig. 4c). The initial and last spindles demonstrated significant increases in the relative power of 30–100 Hz ($P < 0.001$, $P < 0.001$) which was not increased in the preceding background. These spindles demonstrated similar strengths of gamma oscillation which were further increased at seizure onset ($P < 0.001$ for both the initial and last spindles). However, 30–100 Hz frequencies were expressed to different levels in the individual brain regions ($P < 0.001$) (Fig. 5c).

Analysis between stage of experiment and brain region

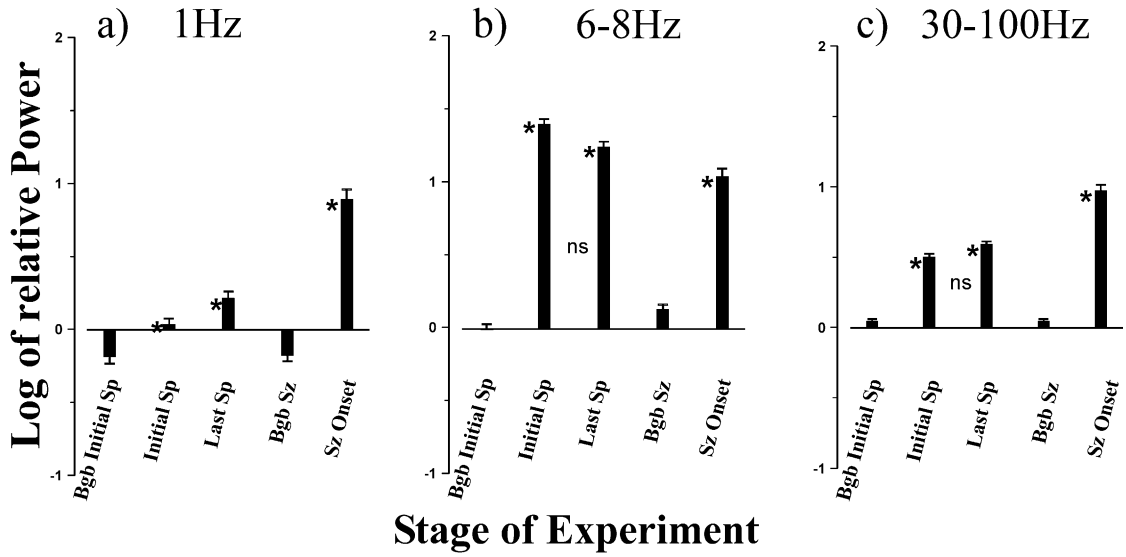

Fig. 4. Means and SEM for all animals from all brain regions demonstrate changes in the relative power of frequencies at time points throughout the experiment for: (a) 1 Hz; (b) 6–8 Hz; and (c) 30–100 Hz. Time points were: background before the initial spindle (Bgb Initial Sp), background before the seizure (Bgb Sz), initial spindle (Initial Sp), last spindle (Last Sp) and the first 2 s of seizure (Sz onset). * = significantly different to baseline levels of power.

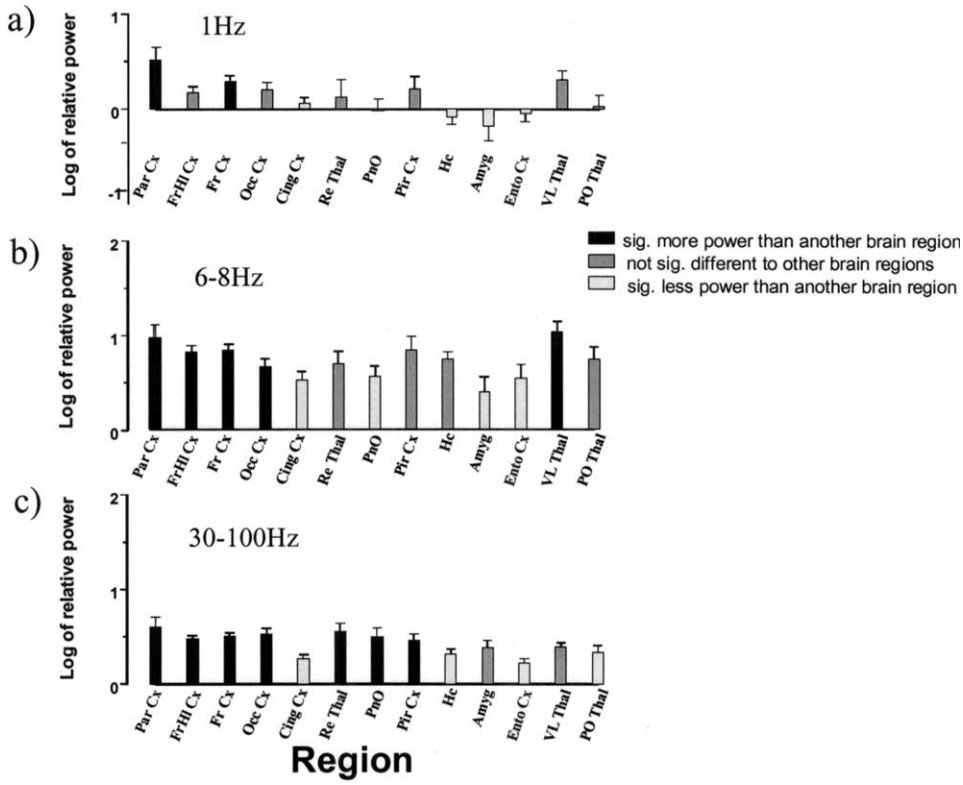

Fig. 5. Means and SEM for all animals from all stages of the experiment demonstrate differential expression of frequencies in various brain regions during epileptogenesis for: (a) 1 Hz; (b) 6–8 Hz; and (c) 30–100 Hz. Abbreviations: Par Cx = parietal cortex, FrHl Cx = frontal hindlimb cortex, Fr Cx = Fr1 area of motor cortex, Occ Cx = occipital cortex, Cing = cingulate cortex, Re Thal = reticular thalamic nucleus, PnO = pontine reticular nucleus-oral, Pir Cx = piriform cortex, Hc = hippocampus, Amyg = amygdala, Ento Cx = entorhinal cortex, VL Thal = ventrolateral thalamus and PO Thal = posterior thalamic nuclear group.

revealed a significant increase in the relative power of gamma frequencies in neocortical areas ($P < 0.001$, Fig. 6a) which was absent from the more primitive cortical and subcortical areas. At seizure onset gamma frequencies were present with significantly more power compared to pre-drug conditions in all brain regions. This increase, however, was only significantly different to the levels detected during the initial and last spindles for frontal hindlimb cortex, frontal cortex, amygdala and pontine reticular nucleus with the power in reticular thalamus and hippocampus differing from only the initial spindles (Fig. 6b). In the neocortex and hippocampus the power of gamma frequencies at seizure onset was increased over the full range of frequencies from 30 to 100 Hz whereas in the reticular thalamus and pontine reticular nucleus it was restricted to around 40 Hz and harmonics thereof (Fig. 7).

Finally, there were significant changes in the power of 1 Hz oscillations throughout the experiment ($P < 0.001$, Fig. 4a) with differential expression in different brain regions ($P < 0.001$, Fig. 5a) and with a significant interaction ($P < 0.001$, Fig. 6a). The power at 1 Hz in the initial spindles differed significantly from preceding background levels ($P = 0.0497$) and further increased during the last spindles ($P = 0.0305$). This was predominantly due to the increases in 1 Hz power in parietal cortex during spindles (Fig. 6a). There was a significant increase in the power of 1 Hz at the beginning of the seizure in every area compared to pre-drug levels, with frontal cortex, frontal hindlimb cortex, reticular thalamus and pontine reticular nucleus demonstrating the largest increases (Fig. 6a).

4. Discussion

This investigation of acute epileptogenesis induced by picrotoxin demonstrates that there are no changes in the non-epileptiform EEG that provide an indication of the emergence of epileptiform activity. This is in contrast to the long lasting changes induced in background EEG after kainic acid treatment (Medvedev et al., 2000). However, investigation confirmed that discharges were generalised and revealed region-specific changes in the nature of the discharge both in terms of spectral power and in the frequencies contributing to the discharges. These EEG changes may have a relationship to rhythms attributable to certain physiological conditions.

4.1. Alpha frequencies

The first discharges to occur during these experiments were identified as probable mu rhythms. This was based

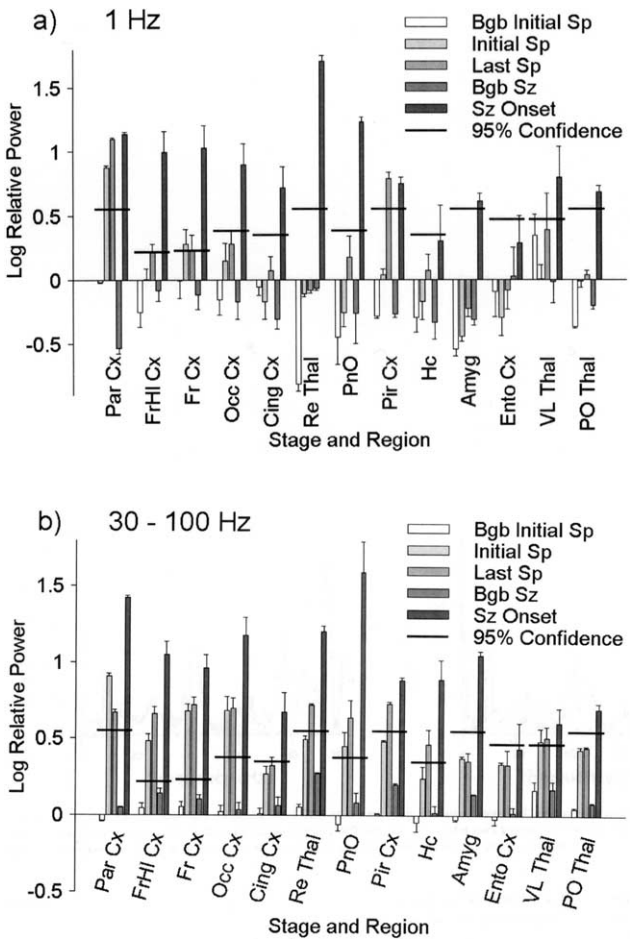

Fig. 6. Means and SEM demonstrate changes in relative power of frequencies in various brain regions for different time points leading to and including seizure onset: (a) 1 Hz; and (b) 30–100 Hz. Abbreviations as per Figs. 4 and 5. The number of animals with particular electrode placements varied for brain regions so the 95% confidence limits for each brain region vary.

on their distribution and frequency and that we have observed them also in untreated animals. They occurred intermixed with picrotoxin-induced spindle discharges indicating that the 12 Hz events and spindles are independent phenomena. They have not been described previously in the rat picrotoxin model. They have been noted in other models of generalised epilepsies but have not been analysed due to their small amplitude failing to meet selection criteria (van Luijtelaar, 1997). These rhythms, generated in the somatosensory cortex, have been shown to be highly coherent with unit firing in the ventral posterior thalamic nucleus (Rougeul-Buser and Buser, 1997) and to be under noradrenergic control (McCormick, 1989). What is unknown is the effect of picrotoxin on single unit activity corresponding to mu activity. However, the unaltered magnitude of the mu rhythm power spectra as the experiment progressed suggests that the rhythms themselves do not appear to contribute to epileptogenesis.

4.2. 6–8Hz frequencies

EEG spindles following picrotoxin, known to be associated with myoclonic jerks, are characterised by a dominant frequency (6–8 Hz) present in all brain regions with maximum power in ventrolateral thalamus and neocortex. The motor phenomenon is likely to be due to spindle-associated excitation in the motor cortex and descending activation of anterior horn cells. This demonstration of 6–8 Hz maximal power increases in ventrolateral thalamus and neocortex is consistent with spindles being predominantly thalamocortical in origin. We hypothesise that the activity at 6–8 Hz described here may be generated by modification of mechanisms shown to be responsible for both sleep spindles (Steriade et al., 1985) and spike and wave spindles during absence epilepsy (Buzsaki et al., 1988; Destexhe et al.,

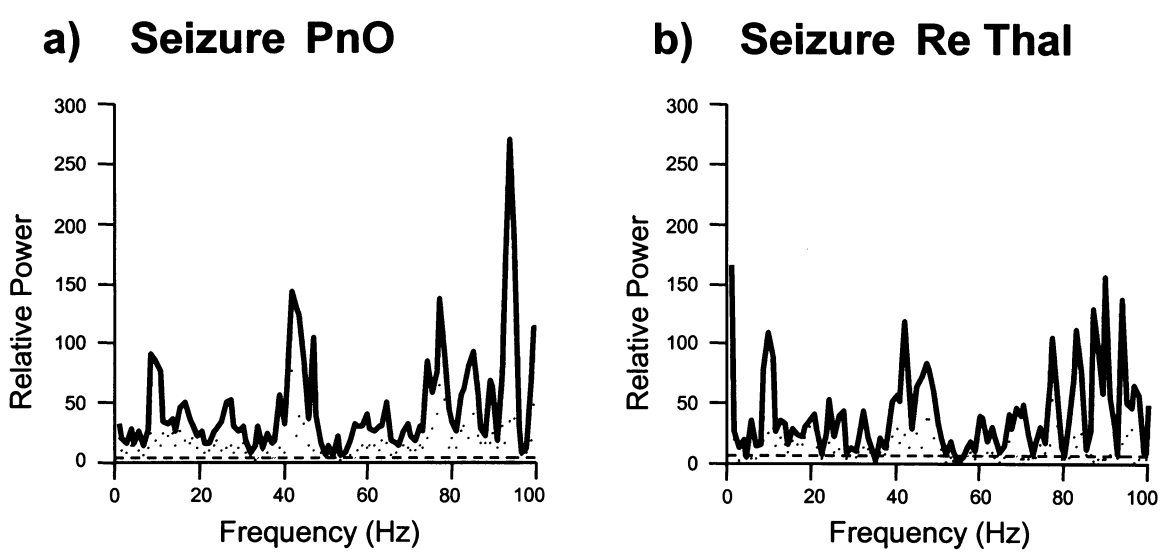

Fig. 7. Examples of average relative responses for segments of EEG at seizure onset displaying increased narrow-band gamma frequencies (around 40 Hz) from: (a) pontine reticular nucleus; and (b) reticular thalamic nucleus, left side only $N = 3$ for both groups. The broken lines indicate the upper and lower confidence limits calculated by F ratio at 0.01 significance level. Dots at each frequency indicate the lower SEM.

1999). Both phenomena depend on thalamic reticular neurons generating oscillatory burst discharges at 6 or 10 Hz under the influence of excitation from the cortex or elsewhere (Steriade and Llinas, 1988; Llinas, 1990; Contreras and Steriade, 1996; Contreras et al., 1997; Destexhe et al., 1998). Following picrotoxin, cortical disinhibition may lead to increased cortical stimulation of these thalamic nuclei. Spindles are also dependent on functional $GABA_A$ receptors in the reticular thalamus for their production. Although picrotoxin affects chloride channels associated with these receptors, it is unknown by how much inhibition is decreased in different brain regions. Chagnac-Amitai et al. have demonstrated that a small decrease in the efficiency of the GABAergic system in neocortex was sufficient to produce synchronised discharges (Chagnac-Amitai and Connors, 1989). We suggest by analogy that when spindles emerge from background EEG numerous thalamic $GABA_A$ receptors remain functional.

From the initial to the last spindle there was a trend for the spectral power of 6–8 Hz to decrease and for the spindles to become more frequent. This slight reduction in 6–8 Hz spectral power may be explained by increasingly disinhibited cortical cells becoming more excited and therefore more involved in the generation of gamma frequencies (see below) and disruptive of spindles. Similarly, at seizure onset the reticular thalamic nucleus also becomes involved in generating gamma frequencies so indicating an increase in excitability of reticular neurons, further reducing the numbers of neurons involved in the 6–8 Hz rhythm. Spindles with their associated myoclonic jerks, which are picrotoxin-induced epileptic events, may well be a model of myoclonic jerks in human PGE. Because there is an increase in the occurrence of spindles towards the timing of the seizure and 57% of the time spindles directly lead into seizure, it is likely that spindles have a role in picrotoxin-induced acute epileptogenesis. Their role may be to lead to metabolic changes throughout the brain and in the neocortex, particularly the motor cortex. Such changes could alter the excitability of the neocortical cells.

4.3. Gamma frequencies

The role of gamma rhythms in the brain is unknown. Rhythmic synchronisation of neural discharges in the gamma band, 30–100 Hz, is thought to provide a spatial and temporal dynamic link between areas involved in the same network. This mechanism has been suggested to solve the so-called 'binding problem' whereby an object, whose parts and properties which are encoded in many distinct brain areas, is experienced as a complete entity (Traub et al., 1997).

Increased high frequency EEG or gamma activity occurred during spindles, mainly in neocortical areas, and during seizures in neocortex and in several subcortical areas. The appearance of gamma first in neocortex is consistent with the suggestion that it is generated in these structures and spreads to the subcortical regions as the process of epileptogenesis progresses. Because the power increase in the gamma range is broad band, it is unlikely to be due to harmonics of the spindle waveform. Its presence in neocortical regions during spindles and absence in subcortical and more primitive cortical structures during spindles suggests that it represents true gamma oscillations and is not due merely to the presence of spindle activity. Faster frequencies are also evident in the raw EEG only from the neocortical regions. Different neuronal networks are capable of generating gamma frequencies. These include pools of excitatory intrinsically oscillatory neurons (Gray and McCormick, 1996), networks of inhibitory neurons (Wang and Buzsaki, 1996) and populations of excitatory and inhibitory neurons (Llinas, 1990; Eeckman and Freeman, 1990; Traub et al., 1997). Synchronous cortical gamma oscillations have been shown to occur in vivo (Brett et al., 1996) as well as in isolated blocks of cortex (Connors, 1984) suggesting that mechanisms for this form of gamma reside within the cortex. Oscillations at 30–45 Hz have been recorded from layer IV GABAergic neocortical neurons (Llinas, 1990). These cells project to layer V pyramidal cells where they presumably impose this rhythm. It has also been shown that due to intrinsic membrane properties the layer V pyramidal cells are capable of generating synchronised oscillations (Silva et al., 1991) consistent with this idea. The fact that gamma frequencies are closely associated with spindles in neocortical areas indicates that cortical gamma oscillations are still capable of being enhanced after picrotoxin treatment, so that the absence of gamma enhancement in the background EEG is not due to disruption of mechanisms for its production. Furthermore, the broad band gamma frequencies demonstrated may point to neuronal synchronisation, albeit not in a narrow bandwidth.

At seizure onset gamma frequencies were increased in all brain regions. Increased gamma frequencies in neocortex and hippocampus spanned a wide range of frequencies between 30 and 100 Hz whereas in the reticular thalamus and pontine reticular nucleus gamma was increased over a much narrower band of frequencies around 40 Hz (Fig. 7). It is generally proposed that seizure onset is a consequence of extreme disinhibition of cortical neurons (Prince, 1985; Yang and Benardo, 1997) and the spectral response of gamma frequencies we observed at seizure onset is consistent with this view. Direct corticofugal inputs to pontine reticular nucleus and reticular thalamus from layer V pyramidal neurons, could mediate the neocortical pattern of activity on these structures. There were also regional variations in the intensity of increase in gamma EEG power that may be explained by differences in the intrinsic oscillatory capabilities of individual brain regions. This is supported by Traub et al. who reported reticular thalamus displaying synchronous activity at frequencies clustered around 40 Hz in contrast to thalamic projection neurons firing over a range of frequencies between 40 and 70 Hz (Traub et al., 1999). This may also apply to the pontine reticular nucleus,

specifically that one preferred frequency of oscillation may be 40 Hz, however its electrophysiology has not been extensively studied.

If neocortical neurons (frontal hindlimb cortex and frontal cortex) underlie the expression of gamma rhythms during spindles and seizures, then other areas with direct corticofugal inputs could have this pattern of activity imposed on them, for example, reticular thalamus and pontine reticular nucleus. However ventrolateral and posterior thalamus and entorhinal cortex, which all have direct inputs from neocortex only have a marginal increase in gamma strength (Fig. 6b). In contrast, amygdala and hippocampus have polysynaptic inputs from neocortex yet have substantial increases in gamma with seizure. Thalamocortical gamma rhythms have been shown to involve the reticular thalamus in much the same way as in the establishment of thalamocortical spindles (Warren et al., 1994). Possibly the direct excitatory input from cortex to ventrolateral and posterior thalamocortical projection neurons may be sufficiently strong to render them unreceptive to the inhibition from the reticular thalamus resulting in relatively little power of gamma frequencies being generated in these structures. It is feasible that ventrolateral and posterior thalamic neurons are vigorously active, but at frequencies higher than can be analysed reliably here, i.e. above 100 Hz.

4.4. 1 Hz frequency

A slow cortical oscillation, (below 1 Hz) has been proposed as a means of synchronising thalamically generated sleep spindles over widespread territories in the intact brain (Steriade et al., 1993). It has also been implicated in grouping sleep rhythms and has been shown to persist in cortex following thalamectomy (Steriade et al., 1993). This attests to its cortical generation, along with the result that it is absent in the thalamus of decorticated animals (Timofeev and Steriade, 1996). All major classes of cortical neurons display this slow oscillation.

In the present experiments the response spectra in some brain regions revealed increased power of frequencies at 1 Hz. With a basic EEG epoch length of 1.024 s (1024 samples) in our analysis, power spectra had a frequency resolution of 1 Hz and a minimum frequency of 1 Hz. We used the spectral power at 1 Hz not only to determine what was happening at that frequency but also to provide an indirect measure of frequencies less than 1 Hz. The rhythm was not present during non-epileptiform activity suggesting that it does not contribute to the process of picrotoxin-induced epileptogenesis. However this rhythm is prominent at seizure onset in frontal hindlimb cortex, frontal cortex, reticular thalamus and pontine reticular nucleus. Based on its distribution in the neocortex we hypothesise that the 1 Hz rhythm present during epileptiform activity may be a modification of the slow sleep oscillation described in detail during sleep by Steriade and colleagues (Steriade et al., 1993). Because the structures exhibiting highest levels of 1 Hz power at seizure onset also demonstrated the highest levels of gamma frequency power, the mechanism of the slow oscillation may play a facilitative role for seizures.

In conclusion, the epileptiform discharges following picrotoxin, both spindles and seizure are confirmed as being generalised. Although after picrotoxin, gamma oscillations occur during neocortical spindles, they are not augmented in the non-epileptiform EEG indicating gamma oscillations do not have a role in picrotoxin epileptogenesis. In fact, using this method of analysis, no frequencies between 1 and 100 Hz appear to contribute to the epileptogenic process following picrotoxin. Further studies using methods derived from non-linear analysis may enable us to determine regional increases in neuronal synchronisation that are essential for the development of the epileptogenic process. The low frequency 6–8 Hz spindle power changes and their regional expression are consistent with spindles being altered thalamocortical physiological rhythms. We suggest that the spindles lead to metabolic changes that alter the excitability of the neocortical cells, cells whose intrinsic capabilities generate the increase in gamma frequencies. The seizure itself appears to be a complex event consisting of gamma frequencies and a cortically generated 1 Hz rhythm. The gamma and 1 Hz distributions suggest that the seizure is a cortical event with downstream effects on other brain regions.

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References

Ben-Ari Y, Tremblay E, Riche D, Ghilini G, Naquet R. Electrographic, clinical and pathological alterations following systemic administration of kainic acid, bicuculline or pentetrazole: metabolic mapping using the deoxyglucose method with special reference to the pathology of epilepsy. Neuroscience 1981;6:1361–1391.

Brett B, Krishnan G, Barth DS. The effects of subcortical lesions on evoked potentials and spontaneous high frequency (gamma-band) oscillating potentials in rat auditory cortex. Brain Res 1996;721:155–166.

Buzsaki G, Bickford RG, Ponomareff G, Thal LJ, Mandel R, Gage FH. Nucleus basalis and thalamic control of neocortical activity in the freely moving rat. J Neurosci 1988;8:4007–4026.

Chagnac-Amitai Y, Connors BW. Horizontal spread of synchronised activity in neocortex and its control by GABA-mediated inhibition. J Neurophysiol 1989;61:747–758.

Coenen AML, van Luijtelaar ELJM. The WAG/Rij rat model for absence epilepsy: age and sex factors. Epilepsy Res 1987;1:297–301.

Connors BW. Initiation of synchronised neuronal bursting in neocortex. Nature 1984;310:685–687.

Contreras D, Steriade M. Spindle oscillation in cats: the role of corticothalamic feedback in a thalamically generated rhythm. J Physiol 1996;490:159–179.

Contreras D, Destexhe A, Steriade M. Spindle oscillations during cortical

spreading depression in naturally sleeping cats. Neuroscience 1997;77: 933–936.

Destexhe A, Contreras D, Steriade M. Mechanisms underlying the synchronizing action of corticothalamic feedback through ingibition of thalamic relay cells. J Neurophysiol 1998;79:999–1016.

Destexhe A, Contreras D, Steriade M. Cortically-induced coherence of a thalamic-generated oscillation. Neuroscience 1999;92:427–443.

Eeckman FH, Freeman WJ. Correlations between unit firing and EEG in the rat olfactory system. Brain Res 1990;528:238–244.

Granata AR. Effects of γ-aminobutyric acid on putative sympatho-excitatory neurons in the rat rostral ventrolateral medulla in vitro. Intracellular study. Neurosci Lett 2001;300:49–53.

Gray CM, McCormick DA. Chattering cells: superficial pyramidal neurons contributing to the generation of synchronous oscillations in the visual cortex. Science 1996;274:109–113.

Hamani C, Mello LEAM. Status epilepticus induced by pilocarpine and picrotoxin. Epilepsy Res 2001;28:73–82.

Inoue M, Peeters BWMM, van Luijtelaar ELJM, Vossen JMH, Coenen AML. Spontaneous occurrence of spike-wave discharges in 5 inbred strains of rats. Physiol Behav 1990;48:199–201.

Kaplan BJ, Williamson PD. Electroencephalogram and somatosensory evoked potential changes after administration of 6 convulsant drugs. Exp Neurol 1978;59:124–136.

King GA. Effects of systemically applied GABA agonists and antagonists on wave-spike ECoG activity in rat. Neuropharmacology 1979;18:47–55.

Llinas R. Intrinsic electrical properties of nerve cells and their role in network oscillation. The brain, NY, USA: The Cold Spring Harbour Laboratory Press, 1990. pp. 933–938.

Massotti M. Electroencephalographic investigations in rabbits of drugs acting at GABA-benzodiazepine-barbituate/picrotoxin receptors complex. Pharmacol Biochem Behav 1985;23:661–670.

McCormick DA. Cholinergic and noradrenergic modulation of thalamocortical processing. Trends Neurosci 1989;12:215–221.

Medvedev A, Mackenzie L, Hiscock JJ, Willoughby JO. Frontal cortex leads other brain structures in generalised spike-and-wave spindles and seizure spikes induced by picrotoxin. Electroenceph clin Neurophysiol 1996;98:157–166.

Medvedev A, Mackenzie L, Hiscock JJ, Willoughby JO. Kainic acid induces distinct types of epileptiform discharge with differential involvement of hippocampus and neocortex. Brain Res Bull 2000;52:89–98.

Newland CF, Cull-Candy SG. On the mechanism of action of picrotoxin on GABA receptor channels in dissociated sympathetic neurones of the rat. J Physiol 1992;447:191–213.

Niedermeyer E. Primary (idiopathic) generalised epilepsy and underlying mechanisms. Clin Electroencephalogr 1996;27:1–21.

Paxinos G, Watson C. The rat brain in stereotaxic coordinates, Sydney: Academic Press, 1986.

Porter RJ. Classification of epileptic seizures and epileptic syndromes. In: Laidlaw J, Richens A, Chadwick D, editors. A textbook of epilepsy, Edinburgh: Churchill Livingstone, 1993. pp. 1–22.

Prince DA. Physiological mechanisms of focal epileptogenesis. Epilepsia 1985;26:S3–S14.

Reutens DC, Berkovic SF, Macdonell RA, Bladin PF. Magnetic stimulation of the brain in generalised epilepsy: reversal of cortical hyperexcitability by anticonvulsants. Ann Neurol 1993;34:351–355.

Rougeul-Buser A, Buser P. Rhythms in the alpha band in cats and their behavioural correlates. Int J Psychophysiol 1997;26:191–293.

Scheffer IE, Berkovic SF. Generalised epilepsy with febrile seizures plus – a genetic disorder with heterogeneous clinical phenotypes. Brain 1997;120:479–490.

Silva LR, Amitai Y, Connors BW. Intrinsic oscillations of neocortex generated by layer 5 pyramidal neurons. Science 1991;251:432–435.

Steriade M, Llinas R. The functional states of the thalamus and the associated neuronal interplay. Physiol Rev 1988;68:649–742.

Steriade M, Deschênes M, Domich L, Mulle C. Abolition of spindle oscillations in thalamic neurons disconnected from nucleus reticularis thalami. J Neurophysiol 1985;54:1473–1497.

Steriade M, Nuñez A, Amzica F. A novel slow (<1 Hz) oscillation of neocortical neurons in vivo: depolarizing and hyperpolarizing components. J Neurosci 1993;13:3252–3265.

Timofeev I, Steriade M. Low frequency rhythms in the thalamus of intact-cortex and decorticated cats. J Neurophysiol 1996;76:4152–4168.

Traub RD, Jefferys JG, Whittington MA. Simulation of gamma rhythms in networks of interneurons and pyramidal cells. J Comput Neurosci 1997;4:141–150.

Traub RD, Jefferys JGR, Whittington MA. Functionally relevant and functionally disruptive (epileptic) synchronised oscillations in brain slices. In: Delgado-Escueta AV, Wilson WA, Olsen RW, Porter RJ, editors. Jasper's basic mechanisms of the epilepsies, Philadelphia: Lippincott Williams & Wilkins, 1999. pp. 709–724.

van Luijtelaar ELJM. Spike-wave discharges and sleep spindles in rats. Acta Neurobiol Exp 1997;57:113–121.

Vergnes M, Marescaux C, Depaulis A, Micheletti G, Warter J-M. Ontogeny of spontaneous petit mal-like seizures in Wistar rats. Dev Brain Res 1986;30:85–87.

Vergnes M, Marescaux C, Depaulis A. Mapping of spontaneous spike and wave discharges in Wistar rats with genetic generalised non-convulsive epilepsy. Brain Res 1990;523:87–91.

Wallace RH, Wang DW, Singh R, Scheffer IE, George Jr AL, Phillips HA, Saar K, Reis A, Johnson EW, Sutherland GR, Berkovic SF, Mulley JC. Febrile seizures and generalised epilepsy associated with a mutation in the Na + -channel beta1 subunit gene SCN1B. Nat Genet 1998;19:366–370.

Wang X, Buzsaki G. Gamma oscillation by synaptic inhibition in a hippocampal interneuronal network model. J Neurosci 1996;16:6402–6413.

Warren RA, Agmon A, Jones EG. Oscillatory synaptic interactions between ventroposterior and reticular neurons in mouse thalamus in vitro. J Neurophysiol 1994;72:1993–2003.

Willoughby JO, Mackenzie L. Non-convulsive electrocorticographic paroxysms (absence epilepsy) in rat strains. Lab Anim Sci 1992;42:551–554.

Willoughby JO, Mackenzie L, Medvedev A, Hiscock JJ. Distribution of Fos-positive neurons in cortical and subcortical structures after picrotoxin-induced convulsions varies with seizure type. Brain Res 1995;683:73–87.

Yang L, Benardo LS. Epileptogenesis following neocortical trauma from two sources of disinhibition. J Neurophysiol 1997;78:2804–2810.